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Annexure II – Mentor Review & Feedback Form

To,
The Organizing Committee
Atomberg Technologies - Atomquest 2025

Team Name: Hack O Giants
Mentor Name & Designation: Dr. Tejas Pramod Naik, Assistant Professor, Department of Mechanical Engineering, COEP Technological University, Pune- 411005, Maharashtra.
Date of Review: 15/10/2025

Assessment

A. Strengths / What is working well

The team has demonstrated commendable enthusiasm, creativity, and teamwork throughout the project. Their understanding of the integration between mechanical and electronic subsystems has been impressive, particularly in the design of the cleaning mechanism and navigation system. The prototype concept is innovative, compact, and energy-efficient, showing a clear grasp of the theme "Development of the House-Cleaning Robot." The students effectively used sensors for obstacle detection and have optimized power usage, reflecting a strong practical orientation and problem-solving mindset.

B. Areas of Improvement

Further refinement is needed in motion control and path optimization to improve the robot's navigation accuracy. The team can also work on improving the suction and cleaning efficiency to ensure better dust collection across varied surfaces. Documentation and simulation validation can be made more detailed to support design justifications. Additionally, time management during the testing phase should be strengthened to allow for more iterative improvements before final submission.

C. Mentor's Inputs / Guidance Provided

I guided the team in the selection of sensor, and material selection for the robot's structure. Regular review sessions were held to help them optimize mechanical design for manufacturability. I also provided insights on presentation structuring and technical documentation to align their work with competition evaluation parameters. The team has shown consistent improvement after each feedback cycle and has demonstrated strong commitment to learning and innovation. Being a mentor, I always motivated my team through every round of the competition encouraging them to think critically, stay focused, and maintain team spirit. Their steady growth and improved technical maturity over successive rounds have been truly rewarding to witness.

Declaration:

This is to confirm that I have reviewed the team's work. In my assessment, I have noted what aspects are working well, where improvements are needed, and how my input supported the team in refining their approach.

Naik 15/10/2025

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